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# Install OpenCV
!pip install opencv-python-headless -q

import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage import data

# Load a sample image
image = data.astronaut()

# Convert RGB to BGR for OpenCV
image_bgr = cv2.cvtColor(image, cv2.COLOR_RGB2BGR)

# Image size
rows, cols = image_bgr.shape[:2]

# Select 3 points from the original image
pts1 = np.float32([[50, 50],
                   [200, 50],
                   [50, 200]])

# Select corresponding points for the transformed image
pts2 = np.float32([[10, 100],
                   [200, 50],
                   [100, 250]])

# Compute the Affine Transformation matrix
M = cv2.getAffineTransform(pts1, pts2)

# Apply Affine Transformation
affine_img = cv2.warpAffine(image_bgr, M, (cols, rows))

# Convert back to RGB for displaying
original_rgb = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2RGB)
affine_rgb = cv2.cvtColor(affine_img, cv2.COLOR_BGR2RGB)

# Display images
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.imshow(original_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(affine_rgb)
plt.title("Affine Transformed Image")
plt.axis("off")
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plt.show()
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Original Image



Affine Transformed Image

